

Sri Vishnu Gopinathan

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Data Engineer with 4+ years of experience designing and orchestrating production grade ETL and ELT pipelines and lakehouse solutions on Azure across industrial, financial services, and research domains. Skilled at building end-to-end data platforms with Azure Data Factory, Databricks, and PySpark from ingesting and migrating terabyte scale datasets into ADLS Gen2, to modeling curated Delta Lake layers through medallion architecture, to exposing governed, analytics ready data via Synapse and Power BI. Combines strong data engineering fundamentals with a deep learning background and CI/CD discipline (Azure DevOps, Terraform, Databricks Asset Bundles) to deliver reliable, scalable data solutions that drive real business decisions.

SKILLS

Languages: Python, SQL, T-SQL, Spark SQL, PySpark, Java, YAML

ML & AI: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, Prophet, XGBoost, Matplotlib, Seaborn, Plotly, MLflow

Data Engineering: Azure Data Factory, Azure Databricks, Delta Lake, Unity Catalog, Medallion Architecture, Azure Data Lake (ADLS Gen2), Azure Synapse Analytics, Databricks Asset Bundles, Self-hosted Integration Runtime, Apache Kafka, SQL Server Agent, ETL/ELT Pipelines

Cloud & DevOps: Azure ML, Azure DevOps, Azure Container Apps, Azure Monitor, Terraform (IaC), Docker, CI/CD Pipelines, GitHub

BI & Visualization: Power BI, Power Query, Power BI Report Builder, DAX, Tableau

Databases: SQL Server, Azure SQL Database, MySQL, MongoDB, Oracle, Microsoft Access

Backend: Flask, SQLAlchemy, PyMongo, REST APIs, Microservices

WORK EXPERIENCE

Data Engineer – Taurus Industrial Group, LLC

Jan 2024 - Present | Port Lavaca, TX

- Migrated more than 1TB of equipment, project and maintenance data from legacy systems into **ADLS Gen2** using **Azure Data Factory** with a **SHIR**. Refined and transformed data from bronze to gold layers through medallion architecture using Databricks notebooks and workflows.
- Modeled the gold layer as a star schema and stored the curated output as **Delta tables** in **ADLS** governed by **Unity Catalog**. Exposed the gold layer as **Azure Synapse** serverless views for downstream analytics and loaded data into **Azure SQL DB** for application consumption.
- Managed source control and deployment of **Databricks** jobs, resources and notebooks through **Databricks Asset Bundles** and **YAML** pipelines in **Azure DevOps CI/CD**.
- Designed Power BI dashboards using **Power Query** and **DAX** to track asset utilization, operational performance, and budget planning decisions. Built paginated reports in Power BI **Report Builder** for PDF and PowerPoint documentation.
- Engineered end-to-end, parameterized Azure Data Factory pipelines integrating data from AS400 to APM (Asset Performance Management) systems via a **self-hosted integration runtime**, reducing manual data transfer effort by **60%** and improving throughput through **SQL query optimization** and incremental loads.
- Built a REST API in **Python Flask** serving badge and expiration data for access validation. Containerized with **Docker** and managed deployment to **Azure Container Apps**.

Summer Associate (Data Scientist) – Navy Federal Credit Union

Jun 2023 - Aug 2023 | Vienna, VA

- Built ingestion pipelines in **Azure Databricks** consuming high volume transactional and operational event streams from **Kafka** topics via scheduled batch reads into **Azure Data Lake**.
- Built **PySpark** processing jobs on Databricks to transform and clean large customer datasets, promoting raw ingested data from bronze to curated silver layers as **Delta tables** within a **medallion architecture**.
- Performed big data analysis of member-level attributes using PySpark to identify key indicators that highly correlate with delinquency in credit card payments and assign members into risk tiers based on the analysis.
- Worked on a hybrid forecasting solution combining **Prophet** with **LSTM** models in **PyTorch** to predict monthly Credit Card Collections returns, contributing to **model tuning and evaluation**.
- Built **Power BI dashboards** combining streaming and batch data sources to deliver risk metrics, delinquency indicators, and customer segmentation insights to collections and risk management teams.

Research Assistant – University of Houston - Victoria

Aug 2022 - Dec 2023 | Victoria, TX

- Collaborated on research analyzing user behavior patterns using web data from Yelp and other review platforms, performing data cleaning and exploratory analysis using **Python** to improve user journey in web applications.
- Worked on collection and preparation of datasets related to student demographic and academic data for predictive analytics, and stored sensitive data securely in **MongoDB** collections with proper encryption.
- Performed **data wrangling** and **analysis** with **Python** and **SQL** to determine the **feature importance** of survey responses in predicting student pass rate, followed by **data preprocessing** and **scaling**.
- Built and performed **hyperparameter tuning** on a **feedforward neural network** for **regression analysis** using the **Keras** and **TensorFlow** libraries in Python to predict student pass rate, helping academic advisors focus more attention on students with lower scores.
- Exposed the model through a REST API using **Python Flask** for real-time inference and built an application as the user interface, supporting model training, validation, and inference.

Data Engineer – Tata Consultancy Services (TCS)

Oct 2020 - Jul 2022 | Bengaluru, India

- Built scalable batch data processing solutions for manufacturing, laboratory, and supply chain datasets from multiple on-prem and cloud sources.
- Developed and maintained end-to-end ETL/ELT pipelines in **Azure Data Factory** for automated ingestion, transformation, and loading of data into **ADLS Gen2**.
- Developed high-performance **PySpark** and **Spark SQL** notebooks in **Databricks** for large-scale data transformation, aggregation, and feature engineering, writing curated data to **Delta Lake**.
- Designed data validation and reconciliation frameworks using **Python** and **SQL**, including row counts, null checks, and schema validation, improving data accuracy and reducing quality issues.
- Developed CI/CD pipelines using **Azure DevOps** and **Terraform (IaC)** to provision data infrastructure and enable seamless code promotion across environments.
- Wrote **SQL stored procedures, triggers, and window functions** to support daily operational reporting.
- Monitored daily batch jobs, investigated failures, and performed reruns to ensure successful data loads.

EDUCATION

University of Houston, Victoria, TX — Master's in Computer Science, GPA 4.0 | Dec 2023

Anna University, India — B.E. in Electronics and Communication Engineering, GPA 3.92 | March 2020

PUBLICATIONS

PatchMSNet: A Multi-Scale Transformer-CNN Hybrid Network for Satellite Image Classification (**IEEE ICIR 2025**); Efficient Multi-Class Analysis of Consumer Complaints Using Frozen MiniLM Embeddings (**IJACSA 2025**); FGAN-AD-U: Unsupervised Multi-Scale Graph-Transformer Reconstruction for Industrial Anomaly Detection (**ICAART 2026**); A Survey of Student Counseling Systems (**AMCIS 2023**).